

The Merton Problem with a Drawdown Constraint on Consumption

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FRE 7821: Optimal Control
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May 2, 2025

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Introduction (Practical Problem to Solve)

- Imagine you are retired, you have accumulated a well amount of wealth over your life.
- You want to find the best way to both invest and consume your wealth in order to maximize your happiness.
- There are some rules:
 - Happiness (utility) is derived solely from consumption.
 - Wealth is essential because it determines possible consumption paths.
 - You are a nice person and you want to leave all your wealth to your children, grandchildren, and so on until eternity.
 - You can never run out of wealth.
 - Once you've reached a certain standard of living, you're unwilling to fall too far below it — you can only scale back consumption to a limited extent. (Can't go from eating caviar every night to bologne)
- How would you do it?

Problem Defined Mathematically

- The investor seeks to maximize the expected infinite horizon discounted utility of consumption by finding the optimal asset allocation and consumption strategies.

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$$V(x, y) = \sup_{(\theta, c)} \mathbb{E} \left[\int_0^\infty e^{-\rho t} U(c_t) dt \right]$$

where U is a CRRA utility function, θ_t is the allocation on the risky asset $dS_t/S_t = \mu dt + \sigma dW_t$, c_t is the consumption, and the wealth process X_t :

$$dX_t^{x, \theta, c} = (rX_t^{x, \theta, c} + \theta_t(\mu - r) - c_t)dt + \sigma\theta_t dW_t$$

$$Y_t^{y, c} = y \vee \sup_{0 \leq s \leq t} c_s.$$

- Our consumption c_t is constrained by a proportion of the running maximum

$$c_t \geq bY_t^{y, c}, \quad b \in (0, 1)$$

Model setup

- x : Initial wealth
- y : Initial running maximum of past consumption
- X_t : Wealth process at time t
- Y_t : Running maximum of consumption up to time t
- c_t : Consumption at time t
- θ_t : Investment in risky asset at time t
- r : Risk-free interest rate
- μ : Expected return of risky asset
- S : Risky stock which follows a GBM process
- σ : Volatility of risky asset
- ρ : CRRA utility discount rate
- R : Relative risk aversion parameter (CRRA utility)
- b : Drawdown parameter as a ratio to past maximum consumption ($0 < b < 1$)

Regularity Conditions

- Let denote $\mathcal{A}_b(x, y)$ the set of investment and consumption strategies (θ, c) which satisfies:
 - ① Squared-integrability on θ_t , $\int_0^T \theta_t^2 dt < \infty$ *a.s.* $\forall T > 0$
 - ② Integrability on c_t , $\int_0^T c_t dt < \infty$ *a.s.* $\forall T > 0$
 - ③ Drawdown constraint: $c_t \geq bY_t^{y,c} \quad \forall t \geq 0$
- These conditions are sufficient to guarantee $X_t^{x,\theta,c} \geq 0$ *a.s.* and define $X_t^{x,\theta,c}$ as an Itô's process

Hamilton-Jacobi-Bellman (HJB) Equation

- We expect the HJB equation for the problem to be

$$\max \left\{ v_y, \sup_{(\theta, c) \in \mathcal{A}_b(x, y)} -\rho v + \mathcal{A}^{\theta, c} v + U(c) \right\} = 0,$$

where $\mathcal{A}^{\theta, c} v := v_x(rX_t^{x, \theta, c} + \theta(\mu - r) - c_t) + \frac{1}{2}\sigma^2\theta^2 v_{xx}$.

- The second term (HJB of classical Merton problem) could be rewritten as

$$-\rho v + rxv_x + \sup_{\theta \in \mathbb{R}} \left[\theta(\mu - r)v_x + \frac{1}{2}\sigma^2\theta^2 v_{xx} \right] + \sup_{c \geq by} [U(c) - cv_x]$$

- Solving these suprema will yield $\hat{\theta}$ and $\hat{c}$ (optimal controls)
- v_y becomes active when there is no marginal cost of increasing consumption
- Otherwise, the HJB optimizes over the drift and volatility of the wealth process and the utility of consumption.

Verification Argument

Theorem

Let v be a $C^2(\mathbb{R}^2)$ function with quadratic growth.

- ① Suppose v is a supersolution of HJB. Then $v \geq V$.
- ② Suppose v is a solution of HJB, and assume there exists a maximizer $(\hat{\theta}, \hat{c})$ such that

- ① $\max \left\{ v_y, -\rho v + v_x \left(rx + \hat{\theta}(\mu - r) - \hat{c} \right) + \frac{1}{2} \sigma^2 (\hat{\theta})^2 v_{xx} + U(\hat{c}) \right\} = 0$
- ② for every initial value (x, y) , there is a unique solution to SDE

$$dX_t^{x, \hat{\theta}, \hat{c}} = (rX_t^{x, \hat{\theta}, \hat{c}} + \hat{\theta}_t(\mu - r) - \hat{c}_t) dt + \sigma \hat{\theta}_t dW_t,$$

$$Y_t^{y, \hat{c}} = y \vee \sup_{0 \leq s \leq t} \hat{c}_s$$

- ③ The processes $\hat{\theta}_t$ and $\hat{c}_t$ are a well-defined control processes in $\mathcal{A}_b(x, y)$.

Then $v = V$, and $(\hat{\theta}_t, \hat{c}_t)$ is an optimal Markov control processes.

Let's try a candidate function v .

Verification Argument Part I

- Let $\lambda = \frac{\mu-r}{\sigma}$ and let v be supersolution of HJB equation, then for **any** (θ, c) :

$$\max \left\{ v_y, -\rho v + v_x (rx + \lambda \sigma \theta - c) + \frac{1}{2} \sigma^2 \theta^2 v_{xx} + U(c) \right\} \leq 0$$

- $dY_t^{y,c} \geq 0$ because consumption is a nondecreasing process.
- By Itô formula,

$$\begin{aligned} d\{e^{-\rho t} v(X_t^{x,\theta,c}, Y_t^{y,c})\} &= e^{-\rho t} \underbrace{\left(-\rho v + v_x (rX_t^{x,\theta,c} + \lambda \sigma \theta - c_t) + \frac{1}{2} \sigma^2 \theta^2 v_{xx} \right)}_{\leq -U(c_t) \text{ by supersolution}} dt \\ &\quad + e^{-\rho t} v_x \sigma \theta dW_t + \underbrace{e^{-\rho t} v_y dY_t^{y,c}}_{\leq 0 \text{ by supersolution}} \\ &\leq -e^{-\rho t} U(c_t) dt + e^{-\rho t} v_x \sigma \theta dW_t \end{aligned}$$

Verification Argument Part I

Note that Y_t had no quadratic variation term, because it is a non-decreasing process. Integrate over $[0, T]$ and take expectation

$$\underbrace{\mathbb{E} \left[e^{-\rho T} v \left(X_T^{x, \theta, c}, Y_T^{y, c} \right) \right]}_{\geq 0 \text{ by definition.}} - v(x, y) \leq -\mathbb{E} \int_0^T e^{-\rho t} U(c_t) dt + \underbrace{\mathbb{E} \int_0^T e^{-\rho t} v_x \sigma \theta dW_t}_{=0 \text{ by localization.}}$$

Finally, send $T \nearrow \infty$.

$$v(x, y) \geq \mathbb{E} \int_0^\infty e^{-\rho t} U(c_t) dt$$

Since the inequality holds for any (θ, c) , also does for the supremum of them.

$$v(x, y) \geq \sup_{\theta, c} \mathbb{E} \int_0^\infty e^{-\rho t} U(c_t) dt$$

Verification Argument Part II

- Suppose v is solution of HJB equation with $(\hat{\theta}, \hat{c})$. We know at least one of the two terms in HJB is equal to zero under optimal controls.
- Similar to the obstacle equation of an American option, there is a region of values (X_t, Y_t) such that we don't want to raise the running maximum, called the *equilibrium region* where $v_y < 0$.
- Then, the second term of HJB reaches zero and we also have that $dY_t = 0$ because we are not pushing Y_t up.
- Secondly, if we are on a region on which we are very wealthy compared to the running maximum of consumption, so that it is in our interest to rise $Y_t^{y, \hat{c}}$ and increase consumption from then on. In this region, $v_y = 0$, but as soon as we increase Y_t , we are pull back to the equilibrium region, so we spend no time on this region. Furthermore, it can be proved that $(X_t^{x, \hat{\theta}, \hat{c}}, Y_t^{y, \hat{c}})$ is within the equilibrium region almost surely for all $t \geq 0$.

Verification Argument Part II

Then $v_y dY_t^{y,\hat{c}} = 0$ in either case. Integrating over $[0, T]$ and taking expectation we obtain

$$\mathbb{E} \left[e^{-\rho T} v \left(X_T^{x,\hat{\theta},\hat{c}}, Y_T^{y,\hat{c}} \right) \right] - v(x, y) = -\mathbb{E} \int_0^T e^{-\rho t} U(\hat{c}_t) dt + \cancel{\mathbb{E} \int_0^T e^{-\rho t} v_x \sigma \theta dW_t}$$

$$v(x, y) = \mathbb{E} \int_0^T e^{-\rho t} U(\hat{c}_t) dt + \mathbb{E} \left[e^{-\rho T} v \left(X_T^{x,\hat{\theta},\hat{c}}, Y_T^{y,\hat{c}} \right) \right]$$

Finally, send $T \nearrow \infty$.

$$v(x, y) = \mathbb{E} \int_0^\infty e^{-\rho t} U(\hat{c}_t) dt + \lim_{T \rightarrow \infty} \mathbb{E} \left[e^{-\rho T} v \left(X_T^{x,\hat{\theta},\hat{c}}, Y_T^{y,\hat{c}} \right) \right]$$

By Lemma 8, Arun (2013), the limit above is equal to zero.

$$v(x, y) = \mathbb{E} \int_0^\infty e^{-\rho t} U(\hat{c}_t) dt.$$

Verification Argument Part II

We resort to Arun (2013), which presents two necessary technical lemmas to complete the verification argument:

- **Part 2.2** Lemma 2: The system of equations $dX_t^{x, \hat{\theta}, \hat{c}}$ and $Y_t^{y, \hat{c}}$ has a unique strong solution for initial conditions $(x, y) \in \mathcal{D}_{b/r}$.
- **Part 2.3** Lemma 1: $\hat{\theta}$ and $\hat{c}$ are Lipschitz on $\mathcal{D}_a$.

Then $v = V$, and $(\hat{\theta}_t, \hat{c}_t)$ is an optimal Markov control processes. $\square$

Solving HJB Equation: Optimal Portfolio

HJB Equation

$$\max \left\{ v_y, -\rho v + rxv_x + \sup_{\theta \in \mathbb{R}} \left[\theta \lambda \sigma v_x + \frac{1}{2} \sigma^2 \theta^2 v_{xx} \right] + \sup_{c \geq by} [U(c) - cv_x] \right\} = 0$$

- Let's assume $v_x > 0$, $v_{xx} < 0$. The maximization over θ can be rearranged as

$$\sup_{\theta \in \mathbb{R}} \left[\frac{1}{2} \sigma^2 v_{xx} \left(\theta + \frac{\lambda}{\sigma} \frac{v_x}{v_{xx}} \right)^2 - \frac{\lambda^2}{2} \frac{v_x^2}{v_{xx}} \right] = -\frac{\lambda^2}{2} \frac{v_x^2}{v_{xx}},$$

$$\hat{\theta}(x, y) = -\frac{\lambda}{\sigma} \frac{v_x(x, y)}{v_{xx}(x, y)}.$$

- Before solving the supremum on c , let's do a convenient change of variable.

Homogeneity of Utility function

- We consider a utility function of the form:

$$U(c) = \frac{c^{1-R}}{1-R}, \quad \text{for } R > 0, R \neq 1.$$

- This utility function exhibits homogeneity of degree $1 - R$:

$$U(\lambda c) = \lambda^{1-R} U(c), \quad \text{for all } \lambda > 0.$$

- It is easy to see that homogeneity applies to value function V also. Then, setting $\lambda = \frac{1}{y}$

$$V\left(\frac{x}{y}, \frac{y}{y}\right) = \left(\frac{1}{y}\right)^{1-R} V(x, y)$$

$$v(z) = \left(\frac{1}{y}\right)^{1-R} V(x, y)$$

$$y^{1-R} v(z) = V(x, y) \quad z := \frac{x}{y}$$

Initial conditions: Intuition

- Deposit initial wealth x into bank to earn risk-free return rate r
- $rx \geq by$
- rx represents the maximum sustainable consumption without depleting initial wealth
- Conversely it represents minimum wealth required to sustain initial consumption
- If consumption starts at rx , $\theta = 0$ i.e no risky-asset investment

Initial conditions

- Maximum sustainable consumption at time t defined as:

$$c_t = rX_t^{x,\theta,c} \geq bY_t^{y,c} \quad \Rightarrow \quad \frac{X_t^{x,\theta,c}}{Y_t^{y,c}} \geq \frac{b}{r}.$$

- Initial wealth and consumption variables must fall within $\mathcal{D}_{b/r}$

$$\mathcal{D}_{b/r} := \left\{ (x, y) \in (\mathbb{R}_+)^2 \mid \frac{x}{y} \geq \frac{b}{r} \right\}$$

- Arun (2013) presented the Proposition 1:

If $(\theta, c) \in \mathcal{A}_b(x, y)$ then $(X_t^{x,\theta,c}, Y_t^{y,c}) \in \mathcal{D}_{b/r}$ almost surely for all $t \geq 0$.

- Rewriting our value function, we get

$$V(x, y) = \sup_{(\theta, c) \in \mathcal{A}_b(x, y)} \mathbb{E} \left[\int_0^\infty e^{-\rho t} U(c_t) dt \right] \quad \forall x, y \in \mathcal{D}_{b/r}$$

Homogeneous Linear Transformation

- Let's compute the derivatives $v_x(x, y)$, $v_y(x, y)$ and $v_{xx}(x, y)$ in terms of $v(z)$.
- $v_x(x, y)$:

$$\begin{aligned}
 \frac{\partial v(x, y)}{\partial x} &= \frac{\partial}{\partial x} (y^{1-R} v(z)) = \frac{\partial}{\partial x} \left[y^{1-R} v \left(\frac{x}{y} \right) \right] \\
 &= y^{1-R} \frac{\partial}{\partial x} \left[v \left(\frac{x}{y} \right) \right] \frac{\partial}{\partial x} \left[\frac{x}{y} \right] \\
 &= y^{1-R} \frac{1}{y^{-1}} v'(z) \\
 v_x(x, y) &= y^{-R} v'(z)
 \end{aligned}$$

- $v_{xx}(x, y)$:

$$v_{xx}(x, y) = y^{-1-R} v''(z)$$

Homogeneous Linear Transformation

- $v_y(x, y)$:

$$\begin{aligned}
 \frac{\partial v(x, y)}{\partial y} &= \frac{\partial}{\partial y}(y^{1-R}v(z)) = \frac{\partial y^{1-R}}{\partial y} \cdot v(z) + y^{1-R} \cdot \frac{\partial}{\partial y} \left[v \left(\frac{x}{y} \right) \right] \frac{\partial}{\partial y} \left[\frac{x}{y} \right] \\
 &= y^{-R}(1-R) + y^{1-R} \frac{-x}{y^2} v'(z) \\
 &= y^{-R}(1-R) - y^{-R} \frac{x}{y} v'(z) \\
 v_y(x, y) &= y^{-R} [(1-R) - zv'(z)]
 \end{aligned}$$

Solving HJB Equation: Optimal consumption

- Recall the HJB equation's right hand side, supremum over c .

$$\sup_{c \geq by} [U(c) - cv_x] \quad , \quad U(c) = \frac{c^{1-R}}{1-R}$$

- Solving for optimal consumption. First order condition:

$$\begin{aligned} 0 &= \frac{\partial}{\partial c} \left(\frac{c^{1-R}}{1-R} - c y^{-R} v'(z) \right) \\ &= c^{-R} - y^{-R} v'(z) \\ c^{-R} &= y^{-R} v'(z) \\ c &= y \cdot [v'(z)]^{-1/R} \end{aligned}$$

- Only when $y \cdot [v'(z)]^{-1/R} \geq by$, otherwise the supremum is by .

Solving HJB Equation: Optimal consumption

$$\begin{aligned} c &= y \cdot [v'(z)]^{-1/R} \geq by \\ [v'(z)]^{-1/R} &\geq b \\ v'(z) &\leq b^{-R} \end{aligned}$$

- Let now define $z_{b^{-R}} := (v')^{-1}(b^{-R})$ the value of z such that $v'(z) = b^{-R}$.

$$\hat{c}(z) = \begin{cases} by & \text{if } b/r \leq z \leq z_{b^{-R}} \\ y \cdot [v'(z)]^{-1/R} & \text{if } z_{b^{-R}} \leq z \end{cases}$$

Solving HJB Equation: Optimal consumption

- We understand that there should be a region of optimal consumption $\hat{c}$ like a plateau where the consumption is exactly equal to the running maximum.
- Define $z_1 := (v')^{-1}(1)$ the value of z such that $v'(z) = 1$.

$$y \cdot [v'(z)]^{-1/R} \Big|_{z=z_1} = y$$

- We add that the our function $\hat{c}$:

$$\hat{c}(z) = \begin{cases} by & \text{if } b/r \leq x/y \leq z_{b-R} \\ y \cdot [v'(z)]^{-1/R} & \text{if } z_{b-R} \leq z \leq z_1 \\ y & \text{if } z_1 \leq z \end{cases}$$

Final Consumption Piece-wise

- Finally, if there is enough wealth, the consumption will briefly rise beyond the running maximum so it immediately affects the ratio z and it falls again to $z = z_a$ so the consumption goes back to the (new) running maximum.

$$\hat{c}(z) = \begin{cases} by & \text{if } b/r \leq z \leq z_{b-R} \\ y \cdot [v'(z)]^{-1/R} & \text{if } z_{b-R} \leq z \leq z_1 \\ y & \text{if } z_1 \leq z \leq z_a \\ x/z_a & \text{if } z_a \leq z < \infty \end{cases}$$

- If $(X_t, Y_t) = (x_a, y_a)$ and $X_{t+dt} = 1.01x_a$, then $Y_{t+dt} = 1.01x_a/z_a = 1.01y_a$, thus Z_{t+dt} remains constant at z_a .
- $\mathcal{D}_a := \{(x, y) \in (\mathbb{R}_+)^2 : x/y \leq z_a\}$ the *equilibrium* region we mentioned before.

Working on the new ODE

- We want to present our HJB equation in terms of this new variable z to reduce dimensionality. (PDE $\Rightarrow$ ODE)
- The result of $\sup_{\theta \in \mathbb{R}} [\theta \lambda \sigma v_x + \frac{1}{2} \sigma^2 \theta^2 v_{xx}]$ is equal to

$$-\frac{\lambda^2}{2} \frac{v_x^2}{v_{xx}} = -\frac{\lambda^2}{2} \frac{(y^{-R} v'(z))^2}{y^{-R-1} v''(z)} = -y^{1-R} \frac{\lambda^2}{2} \frac{v'(z)^2}{v''(z)}$$

$$\hat{\theta}(x, y) = -\frac{\lambda}{\sigma} \frac{v_x(x, y)}{v_{xx}(x, y)} = -y \frac{\lambda}{\sigma} \frac{v'(z)}{v''(z)}.$$

- The second term in the HJB equation, when written in terms of $v(z)$ and using $\hat{\theta}$ and $\hat{c}$ reads as

$$-\rho y^{1-R} v(z) + r x y^{-R} v'(z) - y^{1-R} \frac{\lambda^2}{2} \frac{v'(z)^2}{v''(z)} + U(\hat{c}) - \hat{c} y^{-R} v'(z) = 0$$

- Factoring out y^{1-R} we have

$$-\rho v + r z v' - \frac{\lambda^2}{2} \frac{v'^2}{v''} + \frac{U(\hat{c})}{y^{1-R}} - \frac{\hat{c}}{y} v'(z) = 0$$

Working on the new ODE

- We have a convenient new form of $U(\hat{c})$:

$$U(\hat{c}(z)) = \begin{cases} y^{1-R} \cdot U(b) & \text{if } b/r \leq z \leq z_{b-R} \\ y^{1-R} \cdot \frac{[v'(z)]^{1-1/R}}{1-R} & \text{if } z_{b-R} \leq z \leq z_1 \\ y^{1-R} \cdot U(1) & \text{if } z_1 \leq z \leq z_a \end{cases}$$

- In this way, everything in terms of the ratio z and $v(z)$ gives

$$\begin{aligned} 0 &= -\rho v + rzv' - \frac{\lambda^2}{2} \frac{(v')^2}{v''} + U(b) - bv' & \text{if } b/r \leq z \leq z_{b-R} \\ 0 &= -\rho v + rzv' - \frac{\lambda^2}{2} \frac{(v')^2}{v''} + \tilde{U}(v') & \text{if } z_{b-R} \leq z \leq z_1 \\ 0 &= -\rho v + rzv' - \frac{\lambda^2}{2} \frac{(v')^2}{v''} + U(1) - v' & \text{if } z_1 \leq z \leq z_a \\ 0 &= (1-R)v - zv' & \text{if } z_a \leq z < \infty \end{aligned}$$

- where $\tilde{U}(v') := \frac{(v')^{1-1/R}}{1-R} - (v')^{1-1/R} = -\frac{(v')^{1-1/R}}{1-1/R}$,

Legendre Dual Transformation

- **What is it?** Given a convex function $\psi(\theta)$, consider

$$\eta = \nabla \psi(\theta),$$

which is the normal vector to the supporting tangent hyperplane to ψ at θ . Observe that the map from θ to η is **one-to-one and differentiable**. One can view this as a change of coordinates, and this transformation is referred to as the *Legendre transform*.

- On generic notation we usually have the following dual relationship

$$\theta = \nabla \phi(\eta), \quad \psi(\theta) = \sup_{\eta} \{\theta \cdot \eta - \phi(\eta)\}$$

$$\eta = \nabla \psi(\theta), \quad \phi(\eta) = \sup_{\theta} \{\theta \cdot \eta - \psi(\theta)\}$$

Legendre Dual Transformation

- In our case, $v(z)$ is concave. So we tweak the transformation accordingly:

$$J(s) := \sup_{z > b/r} \{-(sz - v(z))\} = \sup_{z > b/r} \{v(z) - sz\}$$

- Using the Envelope Theorem we have that

$$J'(s) = -z^*(s),$$

where $z^*(s)$ is the optimal choice.

- Furthermore we have that $J''(s) = -z'(s)$
- Now establish the dual relationship

$$v(z) = \inf_{0 < s < \infty} \{J(s) + zs\}$$

- Note that $v'(z) = s^*(z)$, and $v''(z) = s'(z)$.

Applying the dual transform on the ODE

- As usual, the Hessian of J is the Jacobian of the transformation from dual to primal

$$J''(s) = -\frac{dz}{ds}, \quad v''(z) = \frac{ds}{dz}$$

- Then,

$$J''(s) \cdot v''(z) = -\frac{dz}{ds} \frac{ds}{dz} = -1$$

- We also can deduce that

$$J(s) = v(z^*(s)) - v'(z^*(s)) \cdot z^*(s)$$

- TL;DR: Replace on the differential equation the following

$$v(z) = J + -sJ, \quad v'(z) = s, \quad \frac{1}{v''(z)} = -J''$$

ODE on dual space

$$0 = (1 - R)J + RsJ' \quad \text{if } 0 < s \leq a \quad (R1)$$

$$0 = -\rho J + (\rho - r)sJ' + \frac{1}{2}\lambda^2 s^2 J'' + U(1) - s \quad \text{if } a \leq s \leq 1 \quad (R2)$$

$$0 = -\rho J + (\rho - r)sJ' + \frac{1}{2}\lambda^2 s^2 J'' + \tilde{U}(s) \quad \text{if } 1 \leq s \leq b^{-R} \quad (R3)$$

$$0 = -\rho J + (\rho - r)sJ' + \frac{1}{2}\lambda^2 s^2 J'' + U(b) - bs \quad \text{if } b^{-R} \leq s \leq s_{b/r} \quad (R4)$$

- Constant a is the boundary on dual space of the *equilibrium region*.
- $s_{b/r} = v'(b/r) = \infty$

Solving the Dual ODE

Region R1 ($0 < s \leq a$):

- This is a separable first-order ODE.

$$\frac{dJ}{ds} + \frac{1-R}{sR}J = 0$$

$$\int \frac{dJ}{J} = -\frac{1-R}{R} \int \frac{1}{s} ds,$$

$$\ln |J| = \frac{R-1}{R} \ln |s| + A',$$

$$J(s) = A s^{1-1/R}$$

Solving the Dual ODE

Equilibrium region (R2, R3, R4). Homogeneous solution.

- The following three ODEs are identical except for the non-homogeneous part (the one independent of J). So let solve the following ODE.

$$\frac{1}{2}\lambda^2 s^2 J'' + (\rho - r)sJ' - \rho J = 0$$

- We use an Euler–Cauchy (power-law) ansatz $J(s) = s^m$. Then

$$J'(s) = m s^{m-1}, \quad J''(s) = m(m-1) s^{m-2}.$$

- Substituting into the homogeneous ODE gives

$$0 = \frac{1}{2} \lambda^2 s^2 \cdot m(m-1) s^{m-2} + (\rho - r) s \cdot m s^{m-1} - \rho s^m$$

$$0 = \frac{1}{2} \lambda^2 m(m-1) s^m + (\rho - r) s^m - \rho s^m$$

$$0 = s^m \left[\frac{1}{2} \lambda^2 m(m-1) + m(\rho - r) - \rho \right]$$

Solving the Dual ODE

- Hence the characteristic equation is

$$Q_1(m) := \frac{1}{2} \lambda^2 m(m-1) + (\rho - r)m - \rho = 0.$$

- Let $-\alpha < 0 < 1 < \beta$ be its two solutions. The homogeneous solution becomes

$$J_{\text{hom}}(s) = C_1 s^{-\alpha} + C_2 s^{\beta}.$$

Solving the Dual ODE

Particular solutions. Regions R2 ($a \leq s \leq 1$) **and R4** ($b^{-R} \leq s \leq s_{b/r}$).

- The nonhomogeneous part of these equations are linear: $U(1) - s$ and $U(b) - bs$, respectively.
- A simple Ansatz is the linear function $J_{\text{part}}(s) = a_0 + b_0 s$.
- For instance, replacing on the equation for the latter region

$$s[-rb_0 - b] + U(b) - \rho a_0 = 0$$

- We deduced the solution for ($a \leq s \leq 1$) and ($b^{-R} \leq s \leq s_{b/r}$) regions are respectively:

$$J_{\text{part,R2}}(s) = -\frac{1}{r}s + \frac{U(1)}{\rho}, \quad J_{\text{part,R4}}(s) = -\frac{b}{r}s + \frac{U(b)}{\rho}$$

Solving the Dual ODE

Particular solution. Region R3 ($1 \leq s \leq b^{-R}$).

- Nonhomogeneous part reads $\tilde{U}(s) = -\frac{s^{1-1/R}}{1-1/R}$. Define Ansatz $J(s) = As^k$, with $k = 1 - 1/R$.
- Replacing on the equation we end up with

$$1 = A k \underbrace{\left(-\rho + k(\rho - r) + \frac{1}{2}k(k-1) \right)}_{:=\gamma_M}$$

- Then,

$$J_{\text{part,R3}}(s) = \frac{s^k}{k \gamma_M} = \frac{1}{\gamma_M} \tilde{U}(s)$$

Solving the Dual ODE

$$J(s) = \begin{cases} As^{1-1/R} & \text{for } 0 \leq s \leq a \\ Bs^{-\alpha} + Cs^{\beta} - \frac{1}{r}s + \frac{U(1)}{\rho} & \text{for } a \leq s \leq 1 \\ Ds^{-\alpha} + Es^{\beta} + \frac{1}{\gamma_M}\tilde{U}(s) & \text{for } 1 \leq s \leq b^{-R} \\ Fs^{-\alpha} + Gs^{\beta} - \frac{b}{r}s + \frac{U(b)}{\rho} & \text{for } b^{-R} \leq s < \infty \end{cases}$$

- We deduce two boundary conditions. First, we know that $\theta_t \sim 0$ when z close to b/r , that means $v'/v'' \searrow 0$ as $z \searrow b/r$. Hence,

$$sJ''(s) \rightarrow 0 \text{ as } s \rightarrow \infty$$

- Second, $v(b/r) = U(b)/\rho = \inf_s \{J(s) + \frac{b}{r}s\}$,

$$\lim_{s \rightarrow \infty} \left| J(s) + \frac{b}{r}s - \frac{U(b)}{\rho} \right| \rightarrow 0$$

This implies G has to be zero.

- Continuity of J , J' , and J'' allow us to find the rest of constant starting with F, E, D .

Constants of function J

- $C = \frac{b^{1+R(\beta-1)} - 1}{\beta(\alpha + \beta)} \left[\frac{1}{R\gamma_M} (R(\alpha + 1) - 1) - \frac{\alpha + 1}{r} \right]$

- Threshold a is the root between 0 and 1 of

$$Q_2(m) = (\alpha + \beta)(R(\beta - 1) + 1)Cm^\beta - \frac{(\alpha + 1)m}{r} + \frac{\alpha}{\rho}$$

- $A = \frac{a^{-1+1/R}}{\gamma_M} \left[\frac{1}{1 - R} - a \right]$

- $B = \frac{a^\alpha}{(\alpha + \beta)(R(\alpha + 1) - 1)} \left[\frac{\beta}{\rho} + \frac{(1 - \beta)a}{r} \right]$

- $D = B + \frac{1}{\alpha(\alpha + \beta)} \left[\frac{\beta - 1}{r} - \frac{1}{R\gamma_M} (1 + R(\beta - 1)) \right]$

- $E = \frac{b^{1+R(\beta-1)}}{\beta(\alpha + \beta)} \left[\frac{1}{R\gamma_M} (R(\alpha + 1) - 1) - \frac{\alpha + 1}{r} \right]$

- $F = B + \frac{(1 - b^{1-R(\alpha+1)})}{\alpha(\alpha + \beta)} \left[\frac{\beta - 1}{r} - \frac{1}{R\gamma_M} (1 + R(\beta - 1)) \right]$

Back to Primal Space

- To go back we need the fact that $v(z) = \inf_s \{J(s) + zs\}$ which is attained for $s = (J')^{-1}(-z)$.
- Unfortunately, it's not possible to invert J' analytically in all four regions.

Results

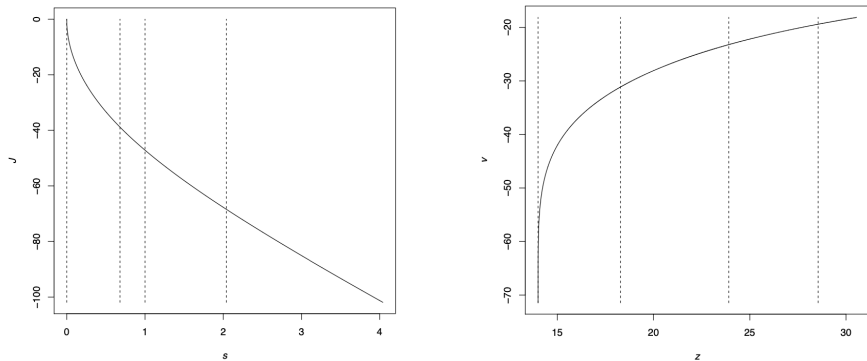


Figure: $J(s)$ on the left. $v(z)$ on the right. The vertical dashed lines represent the critical values b/r , z_{b-R} , z_1 , and z_a which give the boundaries of the four different regions of behavior. For all graphs we take $b = 0.7$, $y = 2$, $R = 2$, $\rho = 0.02$, $r = 0.05$, $\sigma = 0.35$, and $\mu = 0.14$.

Results

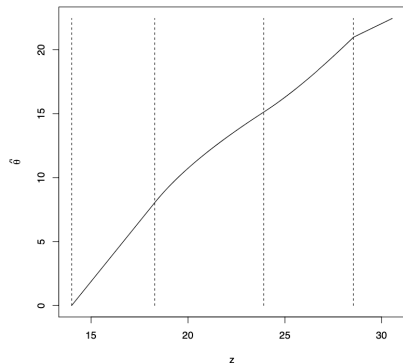
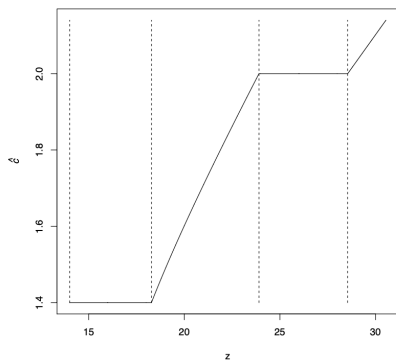


Figure: $\hat{c}$ against z on the left. $\hat{\theta}$ against z on the right.

Results

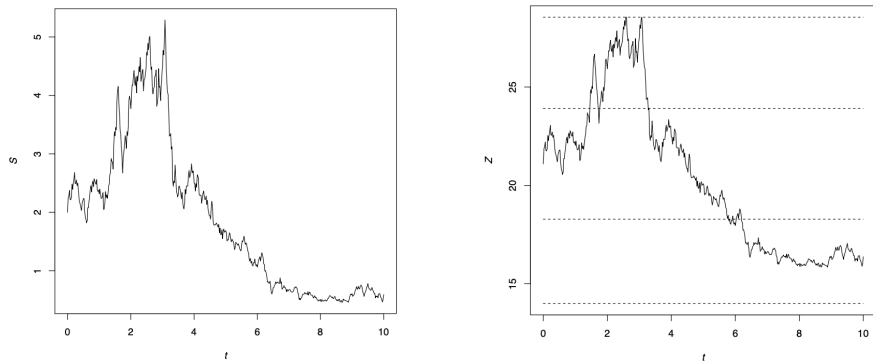


Figure: S_t on the left. Z_t on the right. The above shows a simulation of the stock price and plots of Z against time, t . The horizontal dashed lines represent the critical values b/r , z_{b-R} , z_1 , and z_a which give the boundaries of the four different regions of behavior.

Results

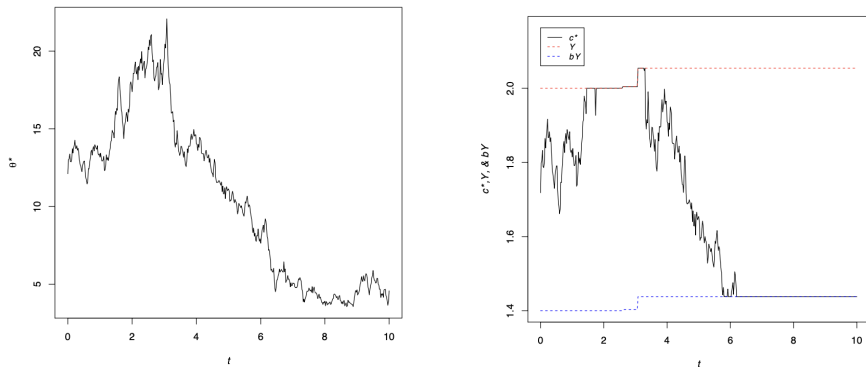


Figure: $\hat{\theta}_t$ on the left. $\hat{c}_t$, Y_t , bY_t on the right.

Conclusion

- **Verification argument & DPE.** Derived the Hamilton–Jacobi–Bellman equation for the drawdown-constrained Merton problem and proved optimality via verification argument theorem.
- **Homogeneity of CRRA utility.** Power-law preferences are *scale-invariant*, letting us collapse the state space from (X_t, Y_t) to the single ratio $Z_t = X_t/Y_t$ and cut the PDE’s dimension by one.
- **Lagrange dual transformation.** Recast non-linear ODE into easy though tedious to solve second-order linear differential equation.
- **Control theory in finance.** Showed how stochastic control tools—DPE, verification—produce economically interpretable rules for consumption and portfolio choice under realistic behavioral frictions.

Reference

- **Arun, T. (2013).** *The Merton problem under a drawdown constraint on consumption.* arXiv preprint. <https://doi.org/10.48550/arXiv.1210.5205>